

# Test #1

## CS5402 — Intro To Data Mining

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### Multiple Choice

1. e. None of the above
2. c. Remove any attribute that has missing values.
3. b.  $\frac{1}{2}$
4. b. wt
5. d. Spearman's rank correlation coefficient
6. c. Healthland
7. b. slice for Time = Q1
8. d. roll up on Location = Beijing or Tokyo (i.e., from city to country)
9. c. drill down on Time = Q1 (i.e., from quarter to month)
10. a. dice for (location = Beijing or Tokyo) and (product = Chain or bracelet) and (time = Q1 or Q2)

### 11 Short Answer

Method #1 is the most accurate, because the true positive ( $y$ -axis) correctly identified the values, while the false positive ( $x$ -axis) incorrectly identified the values. Method #1 had the fastest growing function (with respect to  $y$ ).

## 12 1-R Method

| Attribute     | Attribute Value | # Rows With Attribute Value | Most Frequent Value For sportPref | Errors | Total Errors |
|---------------|-----------------|-----------------------------|-----------------------------------|--------|--------------|
| ageGroup      | youngAdult      | 3                           | football (2)                      | 1      | 3            |
|               | middleAge       | 3                           | football/hockey/baseball (1/1/1)  | 2      |              |
|               | senior          | 2                           | baseball (2)                      | 0      |              |
| gender        | M               | 5                           | baseball/football (2/2)           | 3      | 5            |
|               | F               | 3                           | football/hockey/baseball (1/1/1)  | 2      |              |
| petPreference | dog             | 5                           | football (3)                      | 2      | 3            |
|               | cat             | 3                           | baseball (2)                      | 1      |              |

The rules are as follows:

ageGroup = **youngAdult**  $\implies$  football  
 ageGroup = **middleAge**  $\implies$  football  
 ageGroup = **senior**  $\implies$  baseball

## 13 Prism

For football, we get the following table:

| gender | pet | drink | sport    |
|--------|-----|-------|----------|
| M      | dog | beer  | football |
| F      | dog | beer  | football |

For our P and T values:

|              | T | P | T/P |
|--------------|---|---|-----|
| gender = M   | 3 | 1 | 1/3 |
| gender = F   | 4 | 1 | 1/4 |
| pet = dog    | 3 | 2 | 2/3 |
| drink = beer | 3 | 2 | 3/4 |

Seeing as not T/P values are 1, we must add a clause. We choose pet = dog as the base.

|              | T | P | T/P |
|--------------|---|---|-----|
| gender = M   | 1 | 0 | 0   |
| gender = F   | 1 | 0 | 0   |
| drink = beer | 2 | 2 | 1   |

pet = **dog** and drink = **beer**  $\implies$  football

## 14 Statistical Modeling

The likelihood would be as follows:

$$\text{likelihood} = 4/9 \times 2/9 \times 6/9 \times 3/9 \times 9/14$$

## 15 Entropy

1. entropyBeforeSplit would be as follows:

$$-1/6 \log_2 (1/6) - 2/6 \log_2 (2/6) - 3/6 \log_2 (3/6)$$

2. entropyPoor would be as follows:

$$-2/4 \log_2 (2/4) - 2/4 \log_2 (2/4)$$

3. infoGain would be determined as follows:

$$\begin{aligned} \text{entropyAfterSplit} &= 3/6 \text{entropyShort} + 2/6 \text{entropyMed} + 1/6 \text{entropyLong} \\ \text{infoGain} &= \text{entropyBeforeSplit} - \text{entropyAfterSplit} \end{aligned}$$

## 16 Rule Induction

1. The partitions would be as follows:

$$\begin{aligned} \{d\}^* &= \{\{x_1\}, \{x_2, x_3\}, \{x_5\}, \{x_5\}\} \\ \{e\}^* &= \{\{x_1, x_2, x_5\}, \{x_3, x_4\}\} \\ \{d, e\}^* &= \{\{x_1\}, \{x_2\}, \{x_3\}, \{x_4\}, \{x_5\}\} \end{aligned}$$

2. The coverings are as follows:

- $\{d\}^*$  would not work, because every block in the partition is not a subset of a block in  $\{f\}^*$ .
- $\{d, e\}^*$  would work, because every block in the partition is a subset of a block in  $\{f\}^*$ .
- $\{a, d, e\}^*$  would not work, because although every block in the partition is a subset of a block in  $\{f\}^*$ , it is not minimal.

3. The rules would be as follows:

$$d = X \text{ and } e = 4 \implies f = T$$

$$d = S \text{ and } e = 4 \implies f = T$$

$$d = S \text{ and } e = 3 \implies f = F$$

$$d = H \text{ and } e = 3 \implies f = F$$

$$d = M \text{ and } e = 4 \implies f = F$$

## 17 KD-Tree

Sorting, we get the following:  $[(2, 10), (4, 20), (6, 10), (8, 20), (10, 30)]$ .  
With a median of 6...

- $x < 6$  group:  $[(2, 10), (4, 20)]$
- $x \geq 6$  group:  $[(6, 10), (8, 20), (10, 30)]$

Sorting, we get the following:  $[(2, 10), (4, 20)] [(6, 10), (8, 20), (10, 30)]$   
With a median of 15 for the first group:

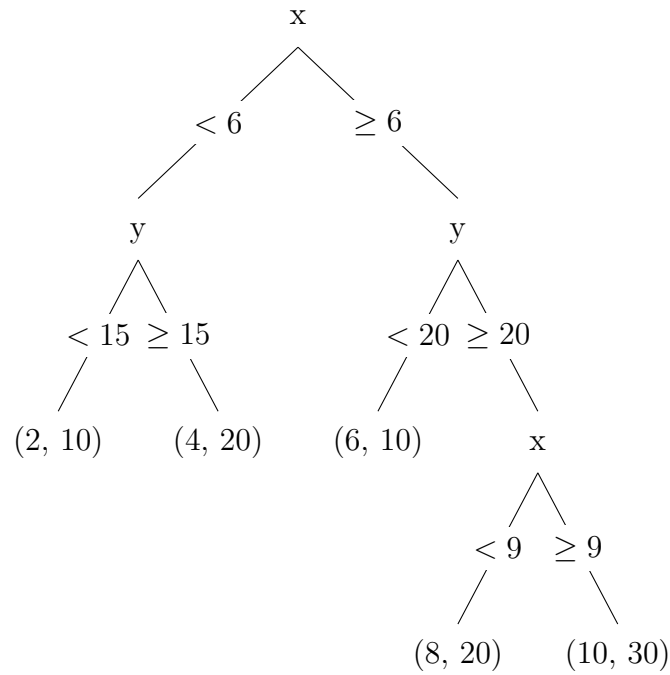
- $y < 15$  group:  $[(2, 10)]$
- $y \geq 15$  group:  $[(4, 20)]$

With a median of 20 for the second group:

- $y < 20$  group:  $(6, 10)$
- $y \geq 20$  group:  $[(8, 20), (10, 30)]$

(Using a shortcut for the final block), Sorting, and using a median of 9, our last block looks like as follows:

- $x < 9$  group:  $[(8, 20)]$
- $y \geq 9$  group:  $[(10, 30)]$



## 18 Clustering

| x | y  | distance to (2, 4) | distance to (5, 6) | distance to (8, 1) |
|---|----|--------------------|--------------------|--------------------|
| 2 | 4  | 0                  | 5                  | 9                  |
| 5 | 6  | 5                  | 0                  | 8                  |
| 8 | 1  | 9                  | 8                  | 0                  |
| 7 | 3  | 6                  | 5                  | 3                  |
| 4 | 10 | 8                  | 5                  | 13                 |
| 3 | 0  | 5                  | 8                  | 6                  |
| 9 | 8  | 11                 | 6                  | 8                  |

Our clusters would be as follows:

**Cluster Center (2, 4)** (2, 4), (3, 0)

**Cluster Center (5, 6)** (5, 6), (4, 10), (9, 8)

**Cluster Center (8, 1)** (8, 1), (7, 3)

With means as follows:

**Cluster Mean of (2, 4), (3, 0)**  $(2.5, 2) \approx (3, 2)$

**Cluster Mean of (5, 6), (4, 10), (9, 8)**  $(6, 8)$

**Cluster Center of (8, 1), (7, 3)**  $(7.5, 2) \approx (8, 2)$

| x | y  | distance to (3, 2) | distance to (6, 8) | distance to (8, 2) |
|---|----|--------------------|--------------------|--------------------|
| 2 | 4  | 3                  | 8                  | 8                  |
| 5 | 6  | 6                  | 3                  | 7                  |
| 8 | 1  | 6                  | 9                  | 1                  |
| 7 | 3  | 5                  | 6                  | 2                  |
| 4 | 10 | 9                  | 4                  | 12                 |
| 3 | 0  | 2                  | 11                 | 7                  |
| 9 | 8  | 12                 | 3                  | 7                  |

**Cluster Center (3, 2)** (2, 4), (3, 0)

**Cluster Center (6, 8)** (5, 6), (4, 10), (9, 8)

**Cluster Center (8, 2)** (8, 1), (7, 3)

*Clusters haven't changed!* Final cluster centers and instances are as follows:

**Cluster Center (3, 2)** (2, 4, 11, yes), (3, 0, 3, yes)

**Cluster Center (6, 8)** (5, 6, 5, no), (4, 10, 8, yes), (9, 8, 1, no)

**Cluster Center (8, 2)** (8, 1, 7, no), (7, 3, 4, yes)

## 19 Confusion Table

1. For a randomly produced results, there were 8 values that we predicted to be B, when they were actually G.
2. For a classifier produced results, there were 30 values that we predicted to be B, and were actually B.
3. The non-random classifier, 90 were predicted correctly. For the random classifier, 39 were predicted correctly. Therefore, 51 more were predicted correctly.
4. Kappa Statistic would be

$$\frac{\text{Non-Random Correct} - \text{Random Correct}}{\text{Total}}$$

Which would be as follows:

$$\frac{90 - 39}{100}$$